



SEP 2025

CHEMICAL ENGINEERING

FOR SUSTAINABILITY AND
COMMERCIAL SUCCESS



ADVANCING CHEMISTRY, MATERIALS
FUNDAMENTALS AND PRODUCTION
PROCESSES

Vikas Berry

Department Head and Satish C. & Asha Saxena Professor

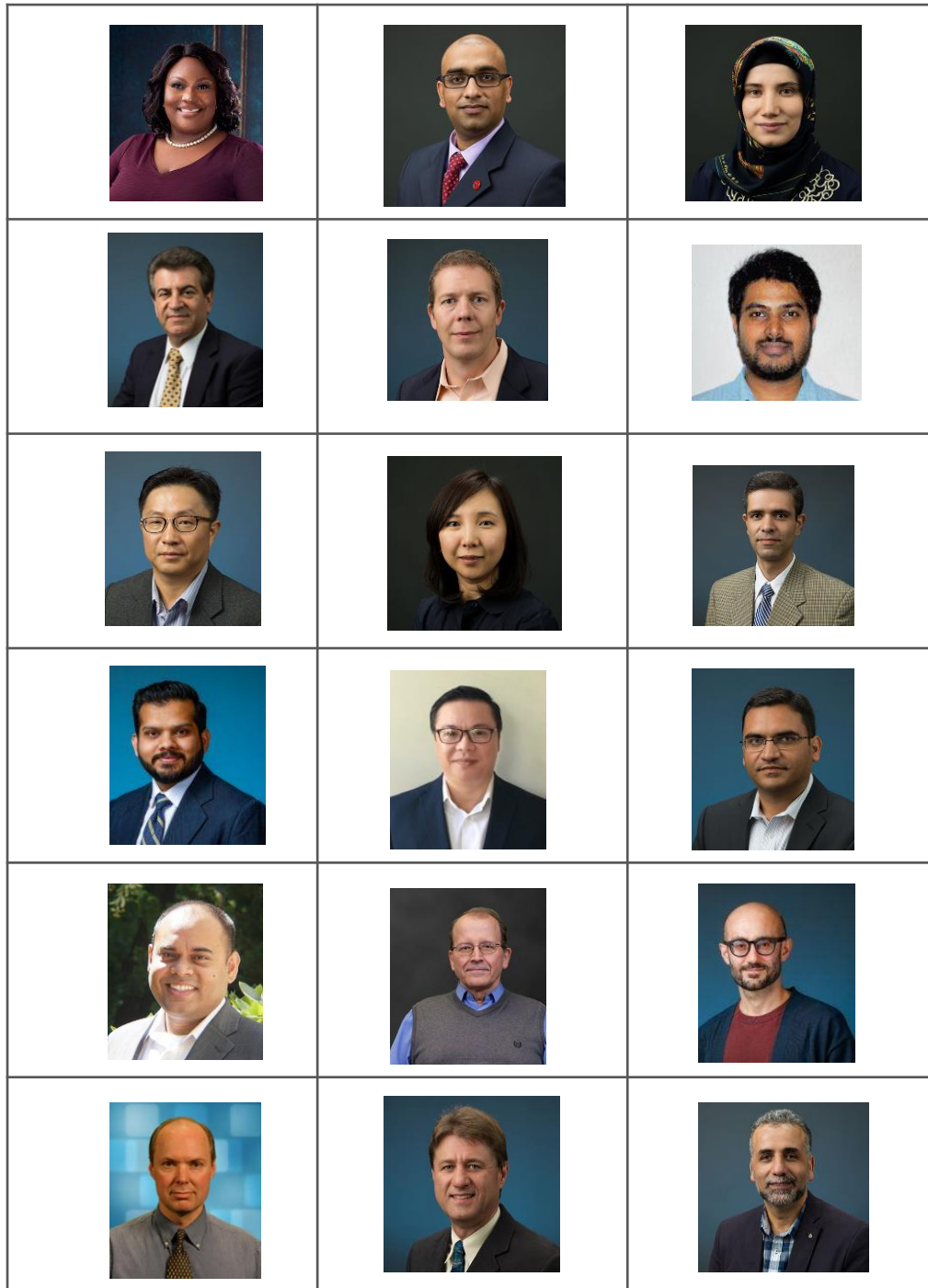


UNIVERSITY OF
ILLINOIS CHICAGO



DEPARTMENT UPDATES





Faculty Members :
18 (13.25 FTE)

National Award Winners //
CEOs // Industry Experts

Tenure-Track : 11

Full : 4
Associate : 5
Assistant : 2
% Female : 18.18%

Clinical-Track: 4

Full : 1
Associate : 2
Assistant : 1
% Female : 25%

Research-Track : 3

Postdoc : 6

Affiliated Faculty : 18



STAFF



Sarai Chavez – Office Support Assistant



Meghan King – Assistant Director of Academic Communications



Eduardo Martinez – Grad./Undergraduate Program Coordinator



Jan Raphael Sagun – Laboratory and Computers Manager

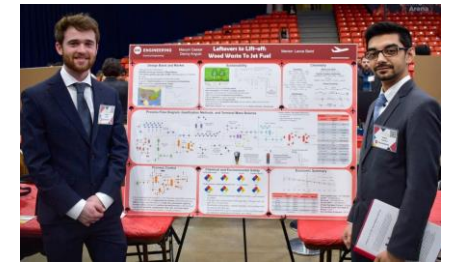
Cassandra Wilson – Business Manager



STUDENTS



BS Degree Program	: 188 students
Graduate Program	: 54 students
MS Degree Program	: 9 students
PhD Degree Program	: 45 students
BS-MS Joint Program	: 5 students



Broadly our story so far

STORY

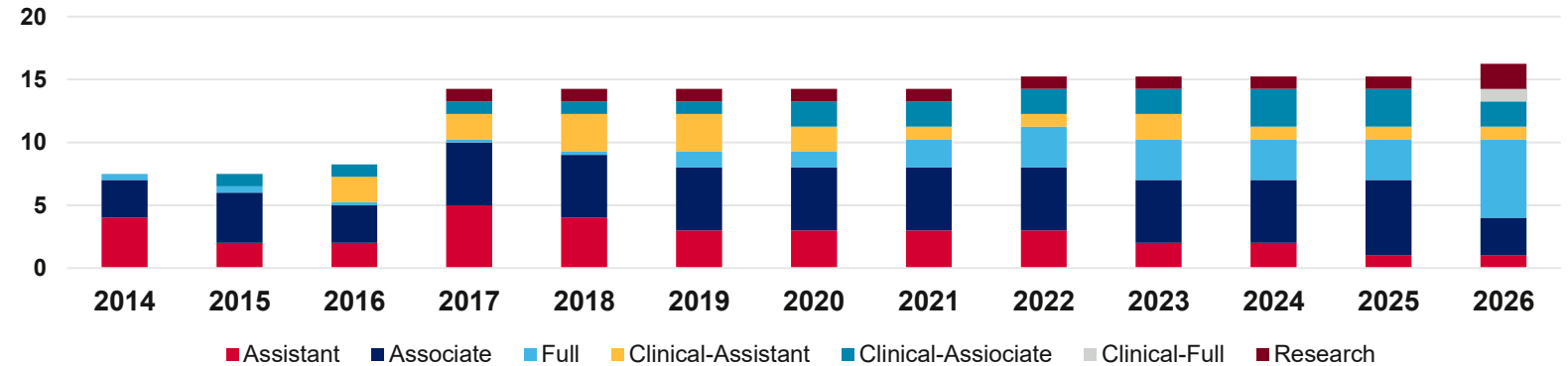


2019

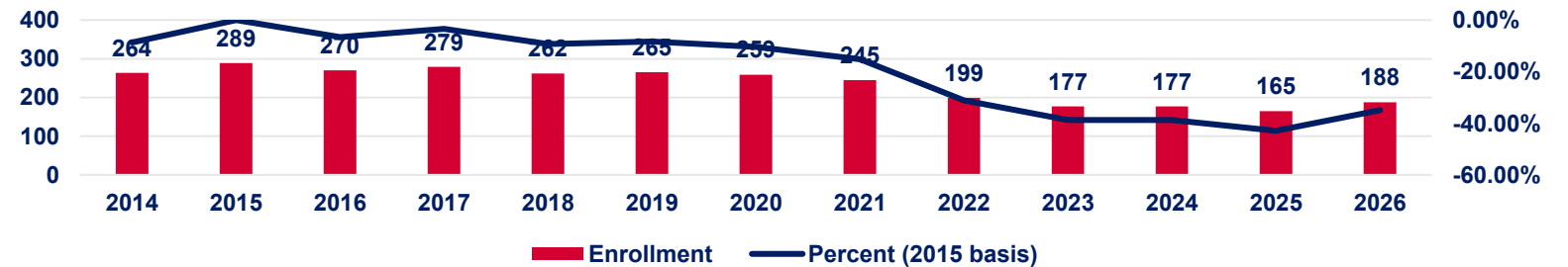


- State-of-the-art research laboratories
- Vibration-damped nanotech labs
- Energy labs: artificial leaf, fuel cells, solar cells, batteries
- Special facilities: bioprocessing lab, dark room, ultrapure water, X-ray spectrometer room
- Enables hands-on student training across new concentration tracks

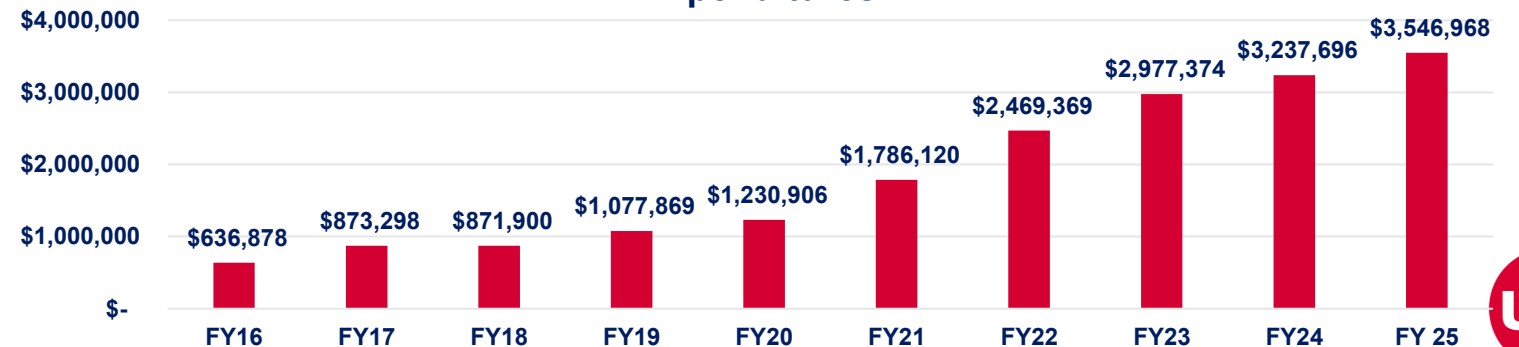
Faculty Size



Enrollment

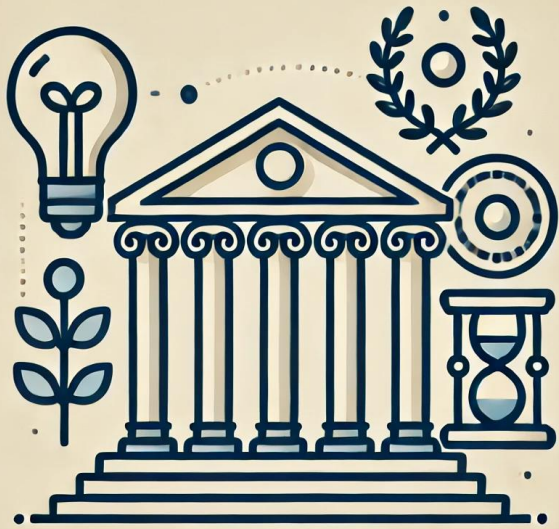


Expenditures



UIC

KEY ELEMENTS



Department Rankings

- **US-NEWS**
 - Chemical Engineering (overall) = 62
 - Chemical Engineering (public) = 38
- **QS Ranking**
 - Chemical Engineering (overall) = 33
 - Chemical Engineering (public) = 20





RESEARCH EXPENDITURES

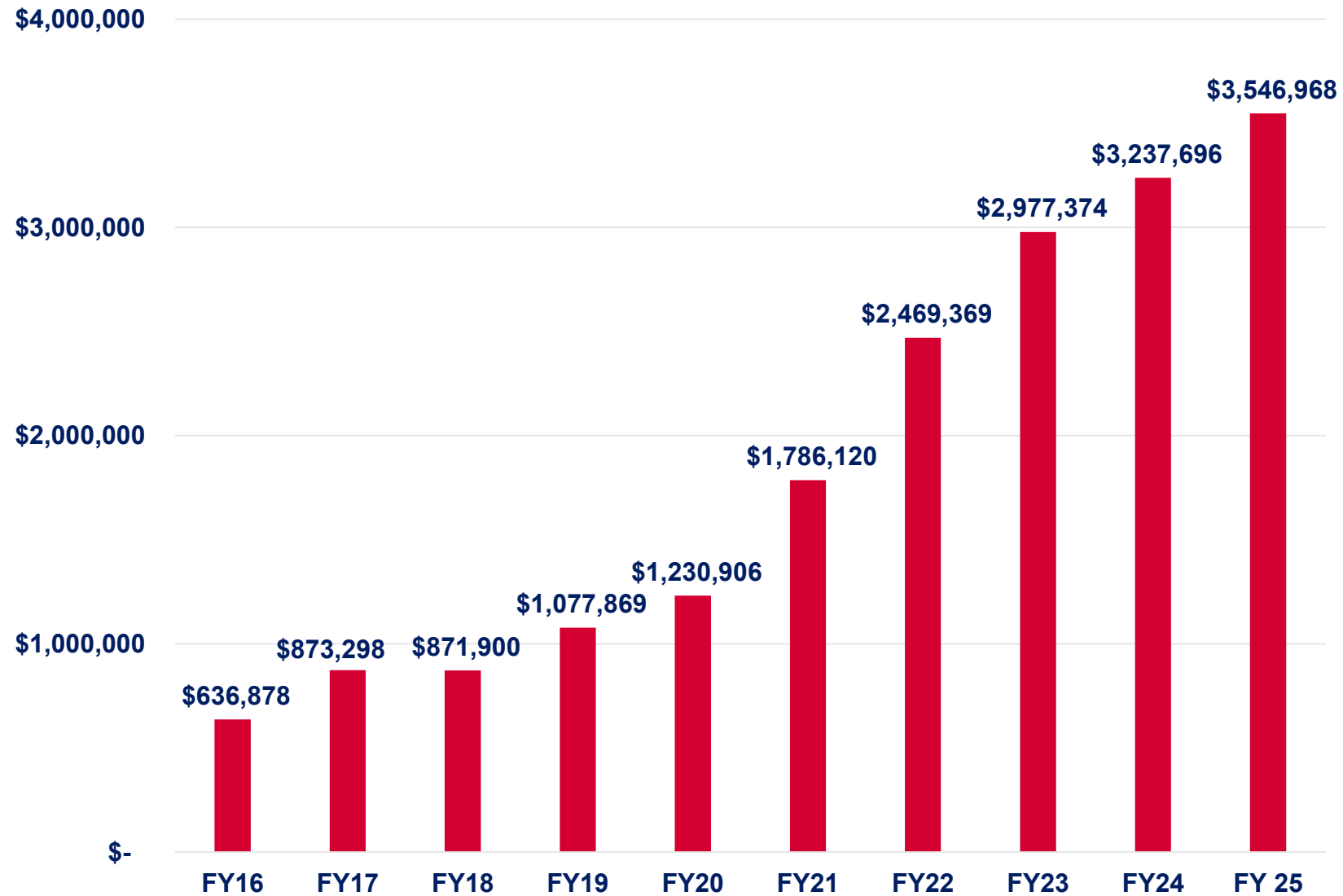


DEPARTMENT UPDATES

Expenditures



Research Expenditures:



Funding in force: \$10,777,121





ENROLLMENTS

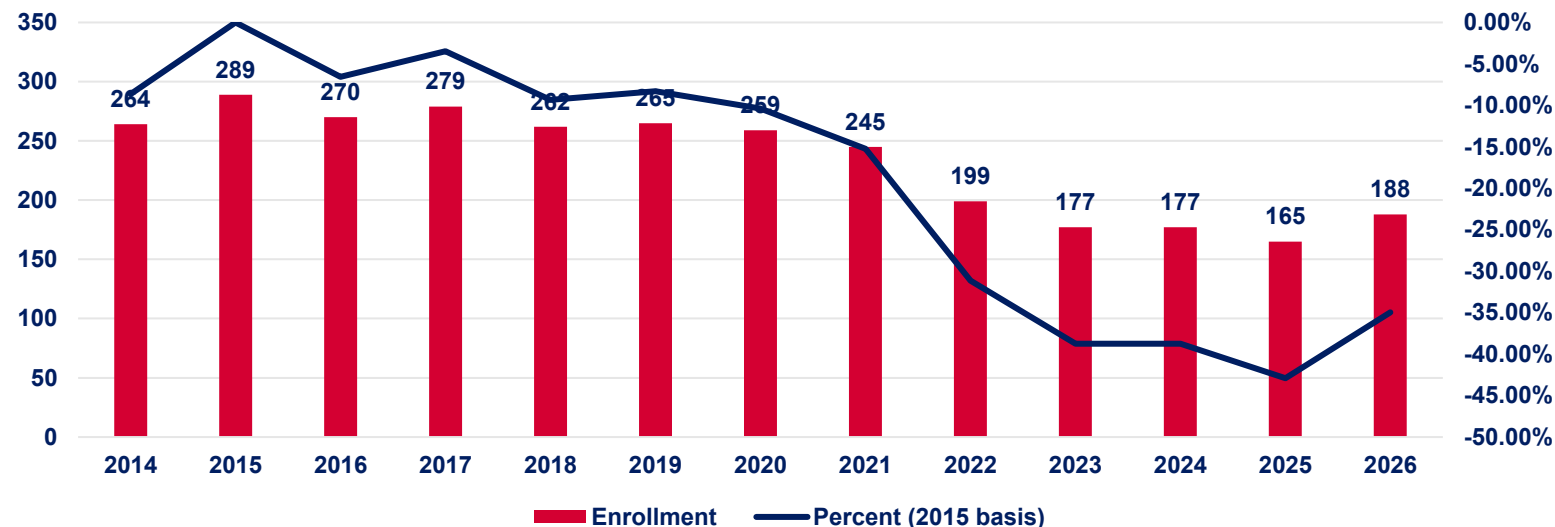


ENROLLMENTS



ENROLLMENTS

Enrollments

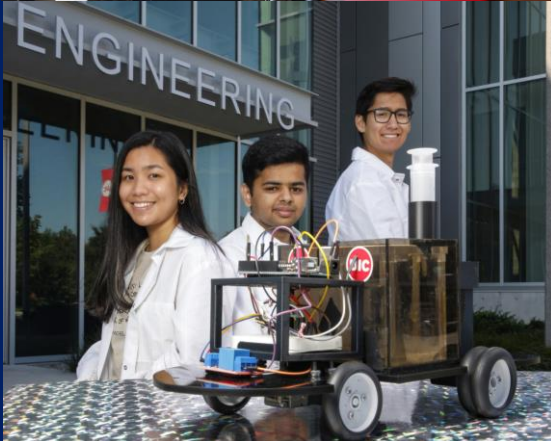


Fall 2025 Enrollments (Based on Last Year's Final #s)

- Freshmen
 - Applications = +33% (COE: 4%)
 - Enrolled = +70% (COE: 15%)
 - 2nd Largest Freshmen Class
- Transfer
 - Applications = +29% (COE: 43%)
 - Enrolled = +33% (COE: 37%)



KEY ACCOMPLISHMENTS



STUDENT PLACEMENTS



L'ORÉAL



Exxon



JACOBS



NALCO

CABOT



GELITA

LA-CO

ECOLAB



Ambitech

GATX



NAVISTAR



SIEMENS

ENEngineering

EXIDE TECHNOLOGIES



South Chicago Packing

Chobani



Air Liquide

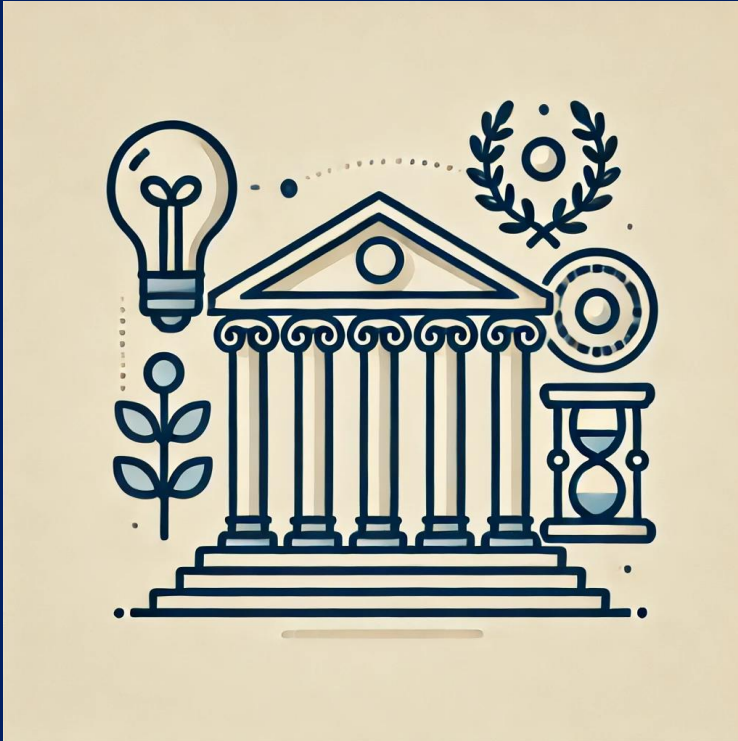


middough





KEY ELEMENTS



Faculty

The Stanford Top 2% researchers' rankings report

Career rankings from 1960:

- Professor Vikas Berry
- Professor Brian Chaplin
- Research Professor Said Al-Hallaj

Citations and publications for 2024:

- Professor Brian Chaplin
- Professor Vikas Berry
- Research Professor Said Al-Hallaj
- Associate Professor Meenesh Singh
- Associate Professor Vivek Sharma

Department Rankings

- US-NEWS
 - Chemical Engineering (overall) = 62
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- QS Ranking
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**WORLD-CLASS
FACULTY MAKES A
WORLD-CLASS
DEPARTMENT**

Entrepreneurship

Chemical Engineering Faculty and Students are collaborating with the following resources to translate ideas into products



UIC - Technology Innovation
Lab

**CHANCELLOR'S
TRANSLATIONAL
RESEARCH INITIATIVE**



13 faculty/student startups
3 CTRI grants received
2 POC grants received
1 DPI grant received
3 NSF-ICORPs
2 operations in mHUB

INITIATIVES



MS in PharmaE

- **Program**
 - 36 credits — thesis / project / coursework
 - Core: Biopharma Process, Particle Tech, Pharm. Characterization
 - Pilot/train: GMP, pilot-scale, characterization
 - Pathway for non-ChE students
- **Industry Value**
 - Workforce-ready (GMP + pilot skills)
 - Local hires — faster onboarding (AbbVie relevance)
 - Research & capstone pipeline for industry problems
- **Board Engagement Opportunities**
 - Course Instruction
 - 1–2 summer paid internships (first cohort)
 - 1–2 capstone problems

UIC Catalyst Center (UIC-CAT)

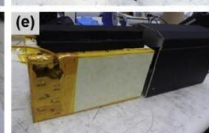
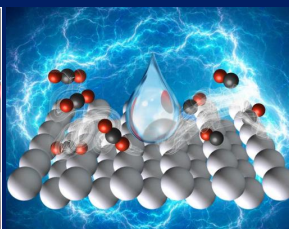
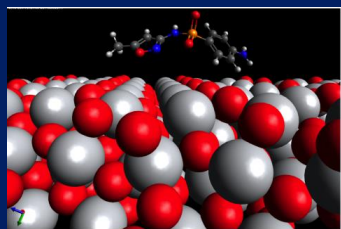
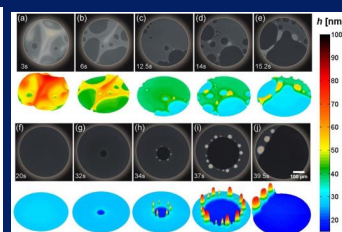
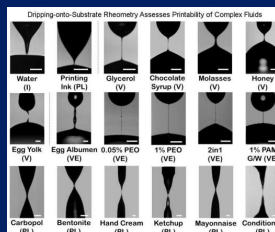
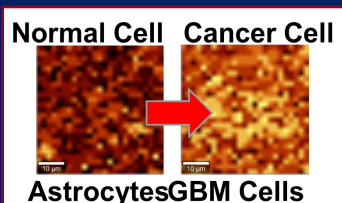
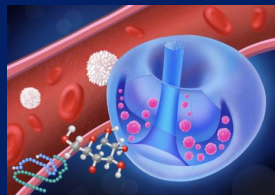
INITIATIVES



- **Mission**
 - Lead in industrial R&D
 - Translate lab innovations to market in collaboration with industry
 - Train industry-ready workforce
- **Capabilities / Thrusts**
 - Photo / Electro / Thermo / Bio-catalysis
 - Reactor design, scale-up & pilot testing
 - High-throughput characterization & AI modeling
 - Shared access: UIC + Argonne facilities
- **Industry value & asks**
 - Value: de-risk tech · talent pipeline · early licensing access
 - Asks: 1) Internships · 2) Seed-sponsored projects · 3) Join Advisory Board / co-develop roadmaps

Chemical Engineering

OTHER OPPORTUNITIES

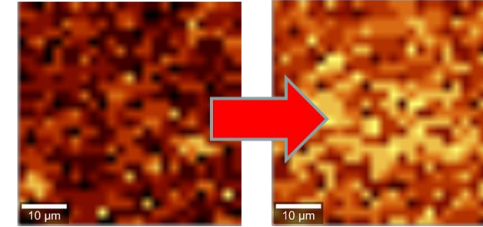


Strategic Research Areas (industry relevance)

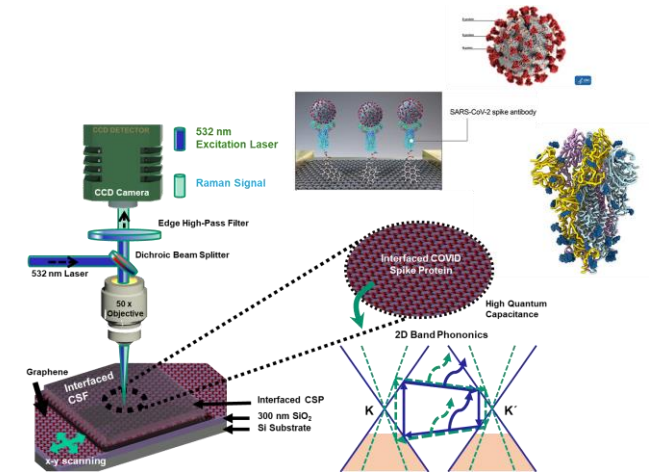
- **Energy** — batteries, fuel cells, solar, artificial leaf (scale-up & materials)
- **Nanotechnology** — advanced materials, coatings, sensors
- **Biomedical Devices** — diagnostics, therapeutics, point-of-care systems
- **Water Science** — contaminant separation & treatment processes
- **AI / Data Science** — process optimization, predictive maintenance
- **Advanced Manufacturing** — pilot-scale prototyping, process intensification
- **Sustainable Foods & Biomass** — plant-based ingredients, conversion tech
- **Complex Fluids & Molecular Engineering** — formulations, separations

- **COVID and Cancer Detection**

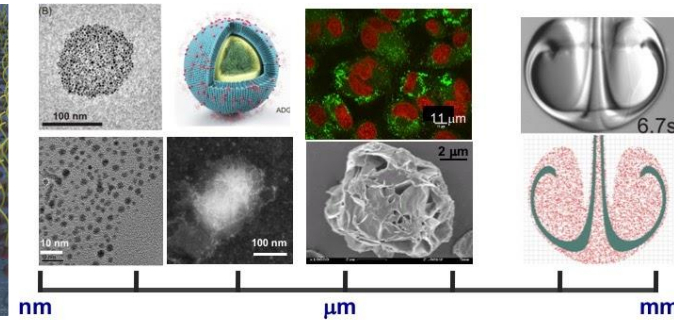
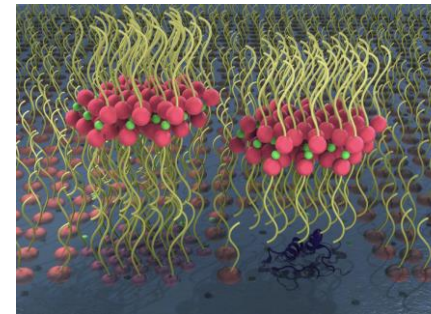
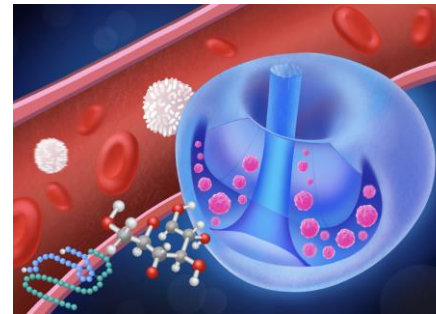
Normal Cell Cancer Cell



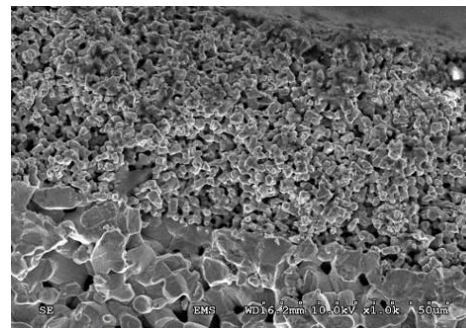
AstrocytesGBM Cells



- **Drug Delivery for Vaccine Development**



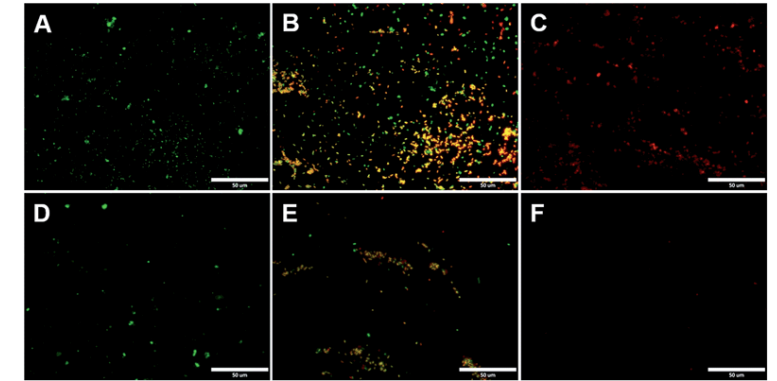
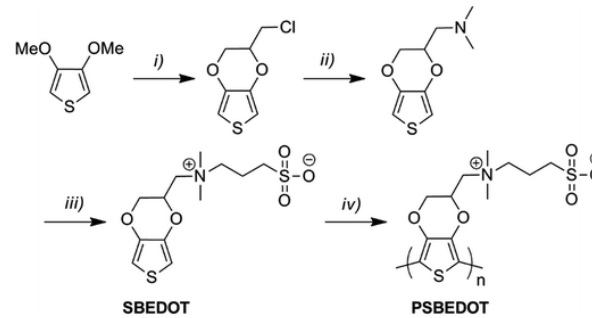
- **Water Purification / NSF ENGINE**



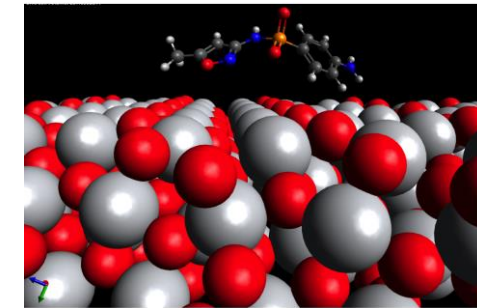
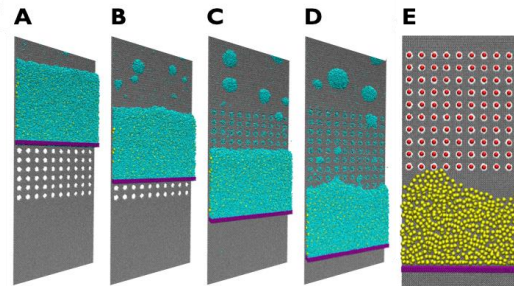
Engineering/Chemical Engineering/

SOME RESEARCH AREAS

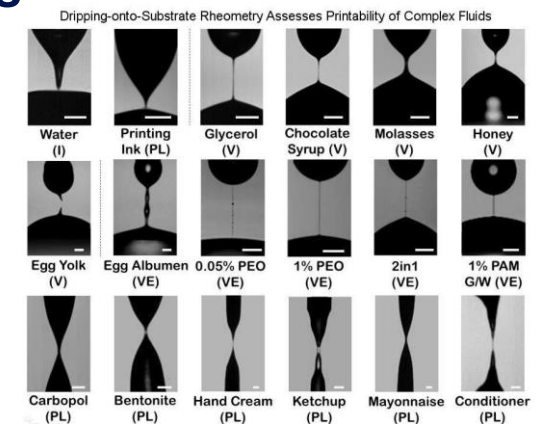
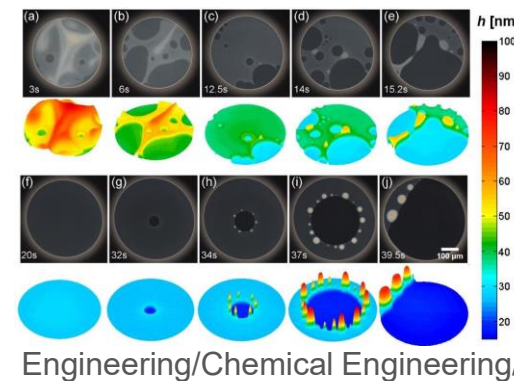
COVID Resistant Polymer Surfaces



Simulation of Self-Assembly in Nanotechnology

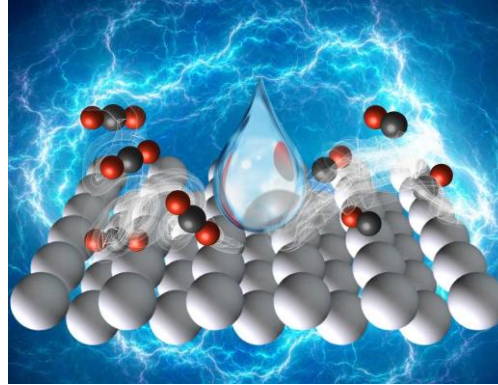


Printability of Complex Fluids

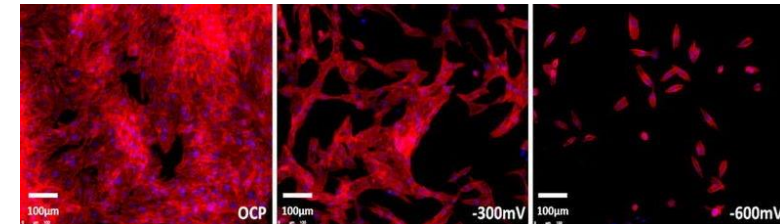


**SOME RESEARCH
AREAS**

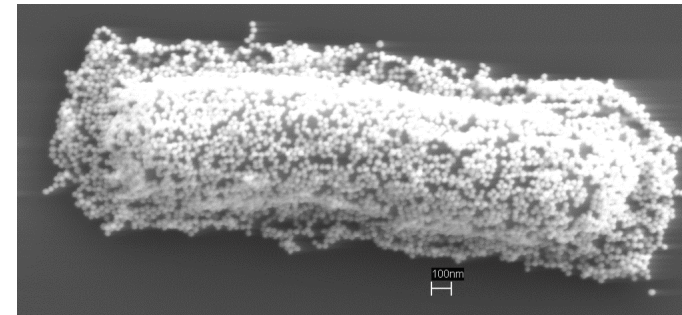
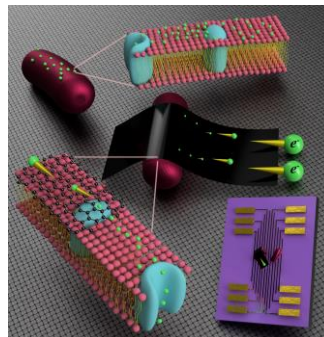
- **Electrocatalysis, Material Chem, Artificial Photosynthesis**



- **Microelectronics Processing**



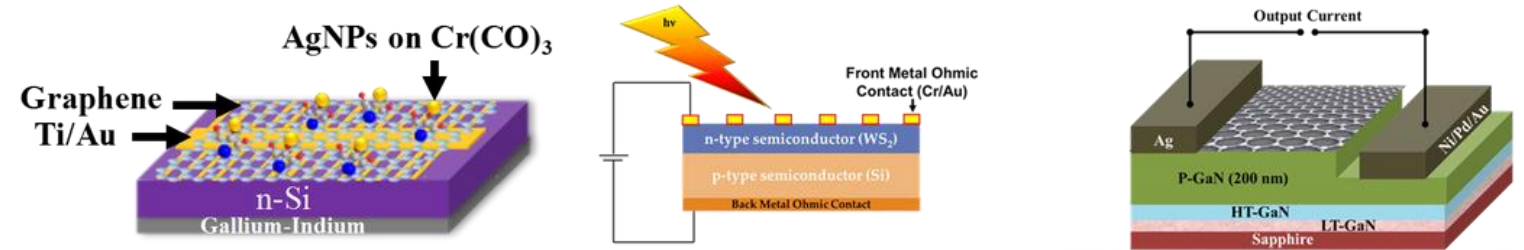
- **Microbial Fuel Cells**



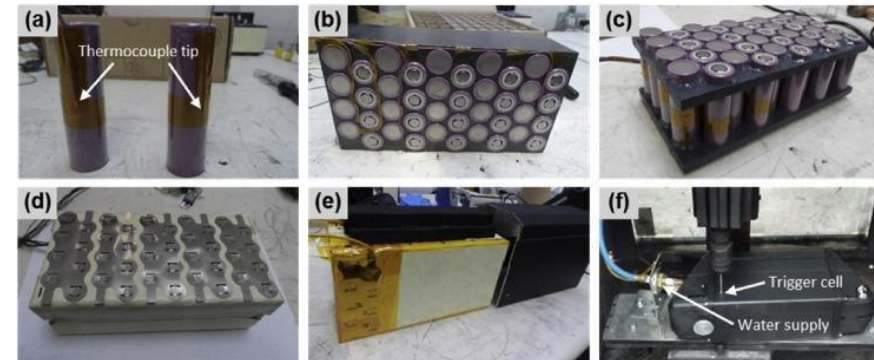
Engineering/Chemical Engineering/

SOME RESEARCH AREAS

- **Few-Atoms-Thick Solar Cells**



- **Thermal Batteries**



- **Bioenergy**



Chemical Engineering at a Glance

18

Faculty

3 NSF Career Awardees

242

Students

\$11M

**FY25 Active
Sponsored Research**

approximately

5

staff

1

**Undergraduate
Degree Programs**

1

Master's Programs

11

Research Labs

1

**Interdisciplinary
Center (coming up)**

QS

Ranking

#20

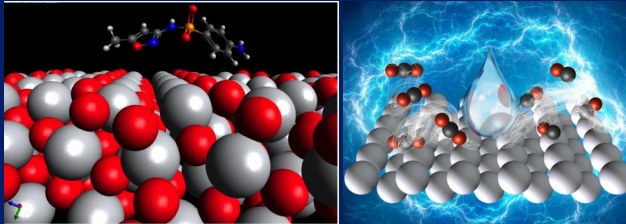
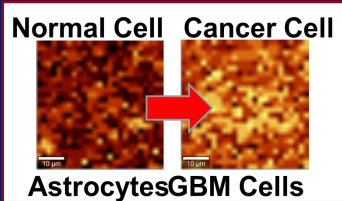
US Public

TOP 33

US

THANKS

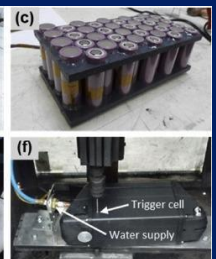
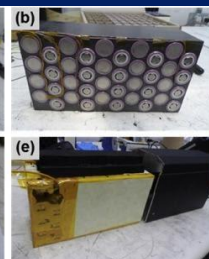
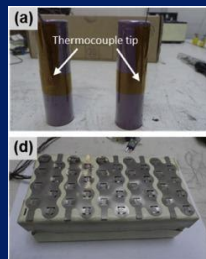
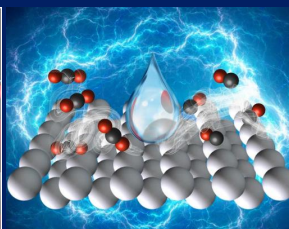
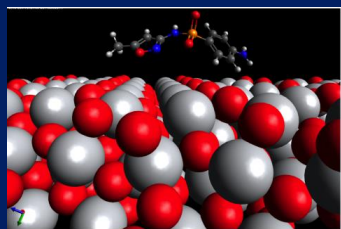
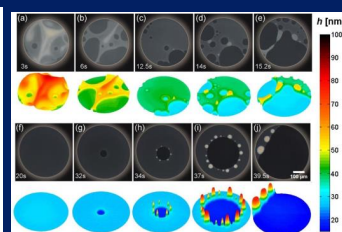
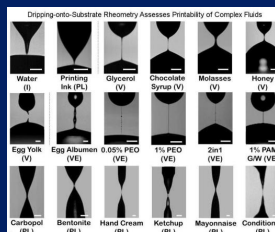
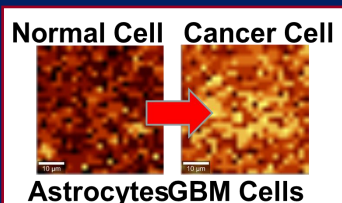
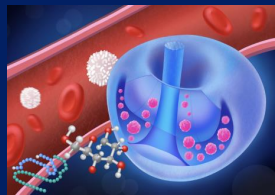




- Endows a chaired professorship with full recurring support for salary, research, and graduate fellowships.
- Premier recognition: named chair, prominent campus visibility, invitation to strategic advisory events.
- Impact: guarantees long-term leadership in a priority area, drives sustained industry-relevant research and workforce development.

Chemical Engineering

ENGAGEMENT OPPORTUNITIES



Collaborate Tier — \$25–75K

Typical commitments:

- **Sponsored seed research project** (\$25–75K, 6–18 months)
- **Co-branded event** (innovation challenge, training workshop)
- **Multi-internship cohort or co-op support** (stipends for 3–5 interns)
- **Large equipment purchase** (unit operations equipment)

Benefits to partner:

- Project milestones and quarterly technical/reporting meetings
- Priority access to student project teams and candidate interviews
- Co-branding on materials + dedicated project page and case study

Partner Tier — \$100K+

Typical commitments:

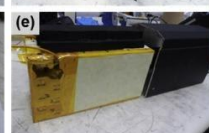
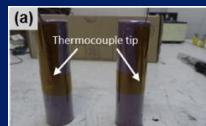
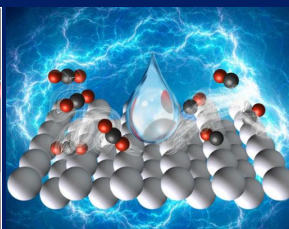
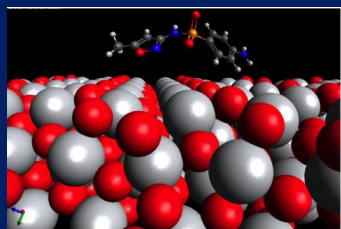
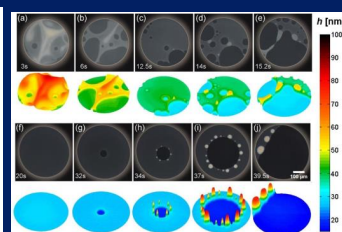
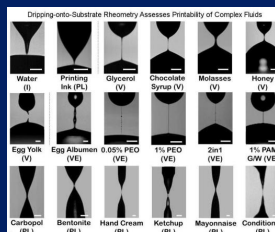
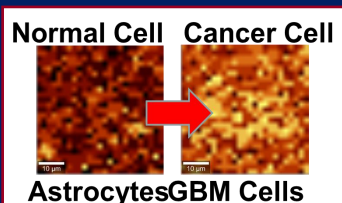
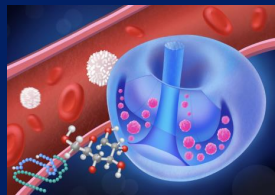
- **Multi-year sponsored research program** (>\$100K/year)
- **Dedicated graduate fellowship** or multi-year internship program
- **Equipment co-funding** for pilot or shared facility
- **Facility renovation / dedicated lab sponsorship**

Benefits to partner:

- Named project or program page, annual impact report with KPIs
- Dedicated faculty liaison
- Customized recruiting events + priority access for hiring

Chemical Engineering

ENGAGEMENT OPPORTUNITIES



In-Kind Tier (Non-cash / Entry)

Typical commitments:

- **Student Mentors**
- **Industry site visits**, plant tours, access to test data
- **Classroom Instruction or Guest Lecture/Workshop**
- **Equipment loan or donation** (benchtop instruments)
- **Small in-kind scholarship** (e.g., travel grants for students to conferences)

Benefits to partner:

- Acknowledgement on website & event materials
- Invitation to department events; priority on student visits

Engage Tier — \$5–10K

Typical commitments:

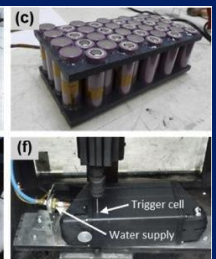
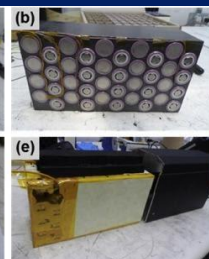
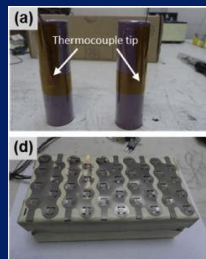
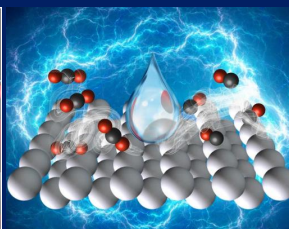
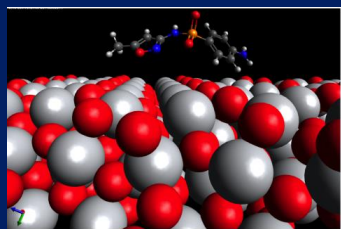
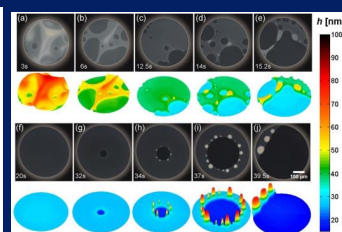
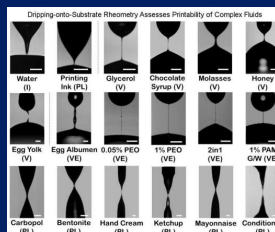
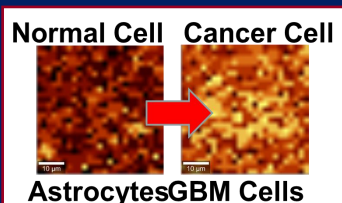
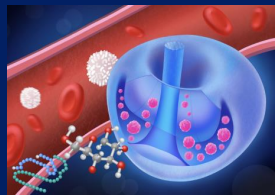
- **Sponsor one senior capstone project** (mentor + \$2k-\$5K materials)
- **Chem-E-Car or Chem-E-Cube support** (\$8k)
- **Equipment donation** (< \$10K)
- **Hire 1–2 paid summer interns** and/or fund an internship stipend
- **Student travel grants** (~\$3k each)
- **Computers** (\$2.5k each)

Benefits to partner:

- Logo on capstone materials + “Engage Tier” listing on site
- Candidate shortlist for internships
- Panel or guest lecture opportunity

Chemical Engineering

OTHER OPPORTUNITIES



Unit Operations Experiments

- Reaction Engineering Experiment: \$40k
- Biochemical Reactor: \$40k
- UV-Vis Spectrometer: \$15k
- Viscosity upgrade: \$5k

Computers

- Equipment: \$2.5k each

Student Travel Grants

- \$3k each

Chem-E-Car

- Equipment+Supplies: \$8k
- Chem-E-Car Fellowships: 3 X \$3k = \$6k

Chem-E-Cube

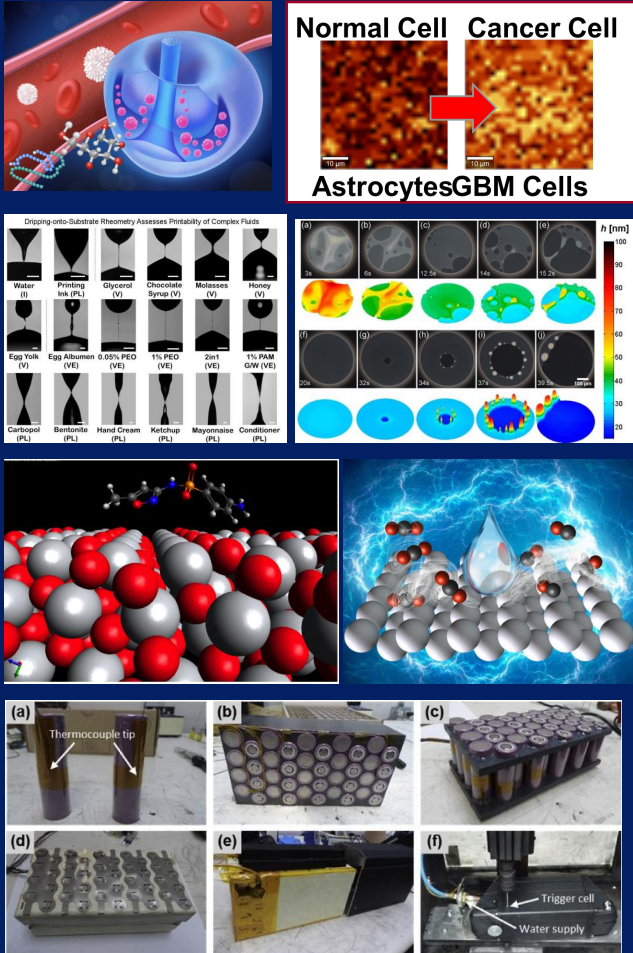
- Equipment+Supplies: \$8k
- Chem-E-Cube Fellowships: 3 X \$3k = \$6k

THANKS



Chemical Engineering

ENGAGEMENT OPPORTUNITIES



Industry Partnership & Talent Opportunities

- **Internships & Co-ops:** paid placements in pharma, energy, materials
- **Research Collaborations:** sponsored projects, SBIR/TEAMING, joint labs
- **Capstone & Problems:** real company problems for senior design teams
- **Faculty Start-ups & Licensing:** early access to spinouts
- **Facility Access:** pilot testing, characterization, prototyping in EIB

How You Can Engage — Concrete Asks

- **Sponsor 1–2 paid interns** next summer
- **Submit 1 capstone problem**
- **Fund a \$50–150K seed project** to de-risk promising tech
- **Join Catalyst Center** (more information coming up)
- **Host a site visit or short industry challenge**

Engineering at a Glance

230*

Faculty
50 NSF Career
Awardees

6,398

Students

6 Departments

13

Undergraduate
Degree Programs

12

Master's Programs

150

Research Labs

4

Interdisciplinary
Centers

\$167M

**FY24 Active
Sponsored Research**

approximately

90

full-time staff

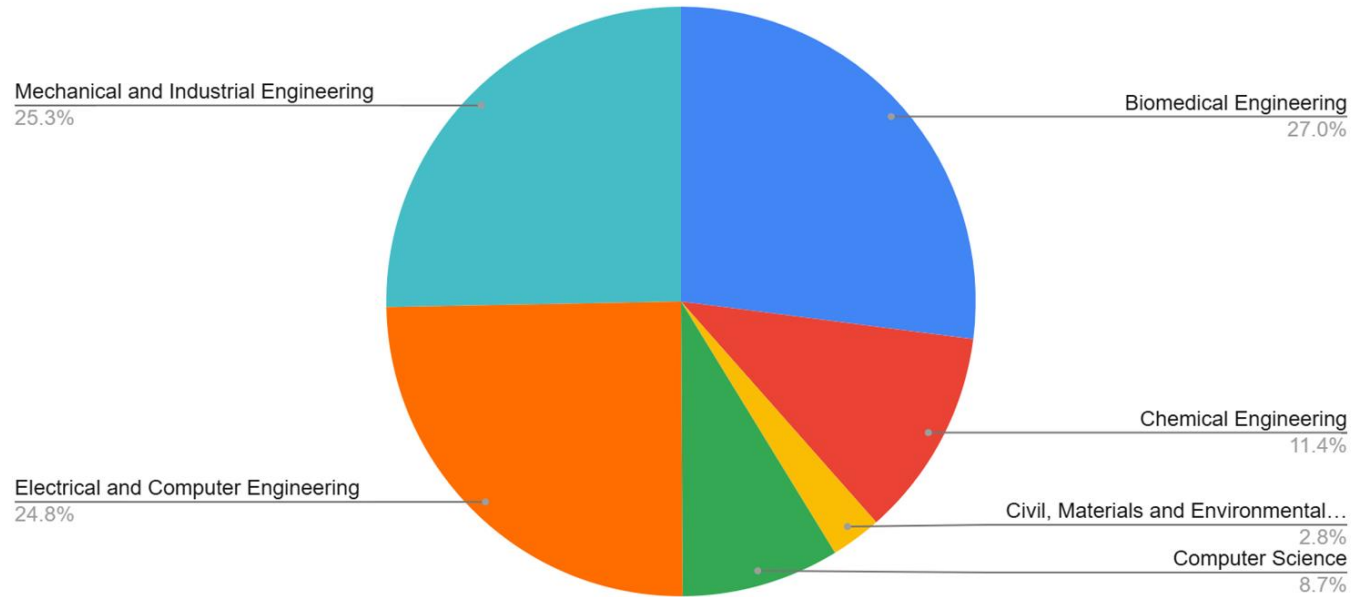
UIC - U.S. News
& World Report
Highlights

#10
Social Mobility

TOP 37

**Ethnically
Diverse**

Engineering Disclosures 2016-2025



Total for COE
577 Disclosures
76 Active Patents
34 Active Licenses

FY	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025*
Disclosures	54	50	57	49	66	62	42	36	45	43
Patents Filed	44	32	38	37	23	37	38	36	38	20
Patents Issued	6	2	8	5	5	7	7	4	5	7
Licenses & Options	6	10	0	5	3	4	1	0	5	2

FY2025 ends 6/30/25

Chemical Engineering Education Overview

- **Fundamentals**
 - Core Principles: Chemistry, Thermodynamics, Heat/Mass-Transfer, Reaction Kinetics
 - Material Science: Structure and properties
 - Fluid Mechanics Principles
 - Mathematical Modeling & Simulation
- **Core Areas**
 - Process Control
 - Process Design & Optimization
 - Separations and Reactions
- **Applications**
 - Chemical Plants & Process Industries
 - Pharmaceutical Design
 - Energy and Oil & Gas
 - Biochemical Engineering
- **Emerging Topics**
 - Sustainability and Green Chemistry
 - Nanotechnology and Materials Innovation
 - Environmental Engineering
- **Professional Development:** Project Management, Safety & Economics
- **Soft Skills**



DEPARTMENT UPDATES

Enrollments

All	2024			2025					
	Apps	Admit	ITE	Apps	% dif	Admit	% dif	ITE	% dif
Chemical Engineering	193	151	51	255	32.12%	196	29.80%	66	29.41%
CME	447	314	114	560	25.28%	384	22.29%	144	26.32%
CS	2894	1701	573	2482	-14.24%	1521	-10.58%	575	0.35%
MIE	1051	731	277	1515	44.15%	1056	44.46%	392	41.52%
Undecided	713	466	178	1011	41.80%	736	57.94%	319	79.21%
COE	7034	4578	1605	7662	8.93%	5138	12.23%	1939	20.81%

Freshmen	2024			2025					
	Apps	Admit	ITE	Apps	% dif	Admit	% dif	ITE	% dif
Chemical Engineering	155	123	29	206	32.90%	163	32.52%	41	41.38%
CME	362	255	69	456	25.97%	324	27.06%	98	42.03%
CS	2476	1466	396	1929	-22.09%	1237	-15.62%	365	-7.83%
MIE	882	607	178	1240	40.59%	881	45.14%	251	41.01%
Undecided	621	382	95	897	44.44%	647	69.37%	230	142.11%
COE	6040	3908	1070	6286	4.07%	4314	10.39%	1281	19.72%

Transfer	2024			2025					
	Apps	Admit	ITE	Apps	% dif	Admit	% dif	ITE	% dif
Chemical Engineering	35	25	19	45	28.57%	30	20.00%	22	15.79%
CME	77	54	42	101	31.17%	58	7.41%	44	4.76%
CS	381	219	163	515	35.17%	267	21.92%	197	20.86%
MIE	158	113	88	257	62.66%	164	45.13%	131	48.86%
Undecided	0	0	0	5	-	0	-	0	-
COE	829	538	408	1184	42.82%	689	28.07%	529	29.66%

Women - All	2024			2025					
	Apps	Admit	ITE	Apps	% dif	Admit	% dif	ITE	% dif
College of Engineering	1819	1372	384	1959	7.70%	1445	5.32%	419	9.11%
Chemical Engineering	81	66	18	110	35.80%	88	33.33%	25	38.89%
Undecided	141	112	26	196	39.01%	151	34.82%	50	92.31%



KEY ELEMENTS

Chemical Engineering is Sustainability

- BS+MS program
- Fast-Track MS program
- 6 Concentrations
 - *Energy and Sustainability*
 - *Molecular Engineering and Polymer Science*
 - *Nanotechnology*
 - *Process Simulations and Automation*
 - *Entrepreneurship and Intellectual Property*
 - *Biochemical Engineering*
- Collaborations
 - LAS, COM, COD, COEd
 - Argonne, Northwestern, U-Chicago, IIT



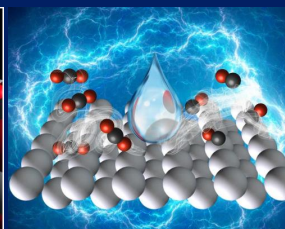
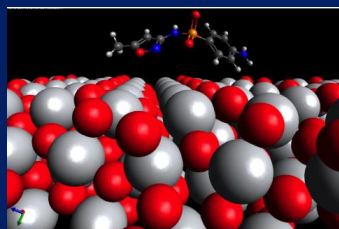
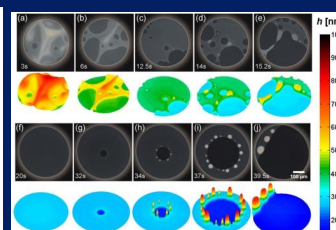
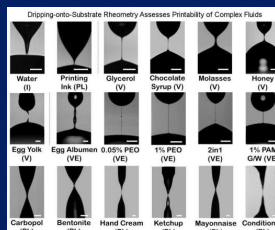
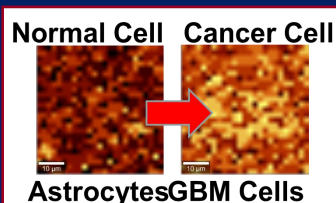
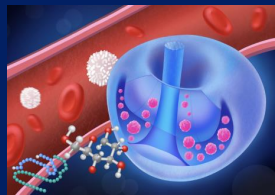
KEY ELEMENTS

Chemical Engineering is Sustainability

- **Electives:**
 - Pharmaceutical Manufacturing Processes
 - Molecular and Macromolecular Engineering
 - Colloidal & Interfacial Phenomena
 - Renewable Energy Technology
 - Fundamentals of Electrochemistry
 - Nanoscale Systems in Chemical engineering
 - Entrepreneurship in Engineering
- **International Programs: China, India and Brazil**
- **Industrial MS program being developed**

Chemical Engineering

ENGAGEMENT



Strategic Research Areas (industry relevance)

- **Energy** — batteries, fuel cells, solar, artificial leaf (scale-up & materials)
- **Nanotechnology** — advanced materials, coatings, sensors
- **Biomedical Devices** — diagnostics, therapeutics, point-of-care systems
- **Water Science** — contaminant separation & treatment processes
- **AI / Data Science** — process optimization, predictive maintenance
- **Advanced Manufacturing** — pilot-scale prototyping, process intensification
- **Sustainable Foods & Biomass** — plant-based ingredients, conversion tech
- **Complex Fluids & Molecular Engineering** — formulations, separations

Industry Partnership & Talent Opportunities

- **Internships & Co-ops:** paid placements in pharma, energy, materials
- **Research Collaborations:** sponsored projects, SBIR/TEAMING, joint labs
- **Capstone & Problems:** real company problems for senior design teams
- **Faculty Start-ups & Licensing:** early access to spinouts
- **Facility Access:** pilot testing, characterization, prototyping in EIB

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